

Yasmine Mustafa Hussain Ahmed Attia

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Summary

PhD Candidate specializing in multimodal medical AI, with expertise in Vision Transformers, Large Language Models, and interpretable deep learning for disease diagnosis. Experienced in integrating neuroimaging (MRI, PET, CT) with clinical and genomic data to enhance diagnostic accuracy and trustworthiness. Strong publication record in top conferences and skilled in both academic research and software engineering. Currently seeking a Fall internship and open to new collaborative research projects.

Education

Missouri University of Science and Technology, PhD Candidate in Computer Science — Rolla, MO Jan 2023 – Expected Jun 2026

Dissertation focus: Multimodality, Explainability, and Trustworthiness of Machine Learning for Disease Diagnosis

- Research on Vision Transformers (ViT) and Large Language Models (LLMs) for multimodal medical data fusion.
- Experiments with CT, MRI, PET, EEG, and genome data, plus tabular data and clinical reports.
- Emphasis on foundation models to improve diagnostic performance and interpretability.

German University in Cairo, M.Sc. in Digital Media Engineering and Technology — Cairo, Egypt Mar 2021 – Jan 2022

Thesis: Brain-Computer Interface for Augmented Reality | Grade: A+

German University in Cairo, B.Sc. in Digital Media Engineering and Technology — Cairo, Egypt Sept 2015 – Jul 2020

GPA: A+ | Semester Abroad: GUC Berlin, Germany (Sept 2019 – Jan 2020), GPA: A+

Bachelor Thesis Abroad: Fraunhofer IGD, Rostock, Germany (Mar 2019 – Aug 2019)

Thesis: Design and Development of a Telemetric Monitoring System for Joint Mobility of Knee or Hip using Machine Learning | Grade: A+

Experience

Full-Time Teaching Assistant, German University in Cairo — Cairo, Egypt Mar 2021 – Jan 2022

- Taught *Data Structures and Algorithms*.
- Taught *Introduction to Computer Programming* (Java).

Full-Time Software Engineer, Everteam Intalio — Cairo, Egypt Feb 2021 – Mar 2021

- Worked with .NET and Microsoft SharePoint.

Part-Time Full Stack Developer, Lirten Solutions — Cairo, Egypt Aug 2020 – Jan 2021

- ReactJS, Node.js, MongoDB, and Postgres development.
- Adapted to new tools and languages within one week; delivered ahead of schedule.

Internships

Deep Learning Intern, ARAtronics Lab (Robotics) — Cairo, Egypt Jul 2018

- Built a deep learning algorithm to control a prosthetic arm using a Myo armband (MATLAB, Neural Networks Toolbox).

Web Developer Intern, German University in Cairo — Cairo, Egypt Jun 2018

- Built a website for the Human++ research cluster (WordPress, PHP, Foundation, jQuery).

Speech Recognition Intern, German University in Cairo — Cairo, Egypt Jul 2017

- Conducted research on speech recognition; collected and edited audio with Audacity.

Publications

1. **Yasmine Mustafa**, & Park, S.-J. (2025). *Anatomical Attention Transformer for Region-Wise Fusion of 3D Imaging and Clinical Text*. Under review at AAAI 2026.
2. **Yasmine Mustafa**, & Park, S.-J. (2025). *Anatomically Informed & Compressed 3D Vision Transformer for Dementia Detection*. Under review at AAAI 2026.
3. **Yasmine Mustafa**, Elmahallawy, M., & Luo, T. (2025). *Unlocking Neural Transparency: Jacobian Maps for Explainable AI in Alzheimer's Detection*. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2025).
4. **Yasmine Mustafa**, Elmahallawy, M., & Luo, T. (2024). *Efficient Brain Imaging Analysis for Alzheimer's and Dementia Detection Using Convolution-Derivative Operations*. IEEE International Conference on Big Data (BigData 2024), pp. 6420–6429. IEEE.
5. **Yasmine Mustafa**, & Luo, T. (2024). *Unmasking Dementia Detection by Masking Input Gradients: A JSM Approach to Model Interpretability and Precision*. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2024), pp. 75–90. Springer Nature Singapore.
6. **Yasmine Mustafa**, & Luo, T. (2023). *Diagnosing Alzheimer's Disease using Early-Late Multimodal Data Fusion with Jacobian Maps*. IEEE International Conference on E-health Networking, Application & Services (Healthcom 2023).
7. **Yasmine Mustafa**, Elmahallawy, M., Luo, T., & Eldawlatly, S. (2023). *A Brain-Computer Interface Augmented Reality Framework with Auto-Adaptive SSVEP Recognition*. Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRINE).
8. **Yasmine Mustafa**, & Eldawlatly, S. (2021). *An Ensemble Classification Approach for Recognizing Steady-State Visually Evoked Potentials Frequencies*. 2021 16th International Conference on Computer Engineering and Systems (ICCES), pp. 1–6. IEEE.

Technologies

Languages: Python, Java, JavaScript, C++ , C, SQL, MATLAB

Technologies: PyTorch, TensorFlow, ReactJS, Node.js, MongoDB, Postgres, .NET, Microsoft SharePoint, MATLAB Neural Networks Toolbox, ANTs, HD-BET

Awards

Best Paper Award — Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2025) Workshop on Pattern Mining and Machine Learning for Bioinformatics 2025

Received Best Paper Award for the work titled "*Unlocking Neural Transparency: Jacobian Maps for Explainable AI in Alzheimer's Detection*".

John W. Claypool Fund for Medical Research Fellowship — Missouri University of Science and Technology 2024

Awarded \$1000 fellowship for research in medical AI.

First Place — 3 Minute Thesis (3MT) Competition — Missouri University of Science and Technology 2024

Won first place in the campus-wide competition and advanced to the regional *Midwestern Association of Graduate Schools (MAGS) 3MT Competition*.